PREM KUMAR SHAW

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PROFESSIONAL SUMMARY

B.Tech Computer Science (Data Science) graduate with strong skills in Python, FastAPI, SQL, backend development, and AI engineering. Experienced in building REST APIs, AI-powered applications, NLP pipelines, machine learning solutions, and data-processing workflows. Hands-on experience with LangChain, LLMs, RAG, Gemini API, databases, and Docker. Strong foundation in DSA, OOP, DBMS, SDLC, and software engineering.

TECHNICAL SKILLS

Languages	:	Python, SQL, Java, JavaScript
AI / GenAI	:	LLMs, LangChain, RAG, Embeddings, Sentence Transformers, NLP
Backend	:	FastAPI, REST APIs, SQLAlchemy, JWT Authentication
Databases	:	MySQL, SQL, MongoDB, PostgreSQL
Web	:	HTML, CSS
ML / Data	:	Scikit-learn, Pandas, NumPy, PySpark, Data Processing
Tools	:	Git, GitHub, Docker, Streamlit, Power BI, VS Code, Jupyter Notebook, Google Colab
Core	:	DSA, OOP, DBMS, SDLC, Operating Systems, Computer Networks, Basic System Design

EXPERIENCE

Forcas Studio Limited | **Data Analytics Intern** *Kolkata - Aug 2025 – Oct 2025*

- Automated daily processing of 800K+ e-commerce records using Python, improving pipeline efficiency by 50%.
- Performed ETL operations and developed SQL queries for data extraction, validation, transformation, and analysis.
- Managed data workflows across Amazon, Flipkart, Ajio, and Meesho e-commerce platforms.
- Built Excel dashboards and Pivot Tables to support business reporting and data-driven decisions.

Techno Exponent | **AI/ML Engineer Intern** *Remote - Aug 2025 – Nov 2025*

- Built and optimized machine learning models for structured datasets using Python and Scikit-learn.
- Developed a CatBoost regression model with feature engineering for diamond price prediction, achieving $R^2 \approx 0.97$.
- Developed prediction logic and deployed the ML application using Streamlit.

Brainware University | **AI/ML Researcher** – *Part-time* *Barasat - Feb 2025 – Sep 2025*

- Developed a **diabetes risk prediction model** using **Python, Machine Learning, Logistic Regression, and Random Forest** on health-indicator data.
- Performed **data preprocessing, exploratory data analysis, statistical analysis, feature evaluation, and model performance evaluation**.
- Applied **AI/ML techniques** to develop an early-detection risk assessment approach for diabetes prevention.
- Used **Pandas, NumPy, and Scikit-learn** for data processing, feature engineering, model training, and evaluation.
- Conducted research and experimentation to compare ML approaches and improve **prediction performance and model reliability**.

PROJECTS

AI Resume Screening Agent | *Python, NLP, Sentence Transformers, Streamlit, Scikit-learn, LLM API* — Live Demo | GitHub

- Built an AI-powered resume screening system that evaluates multiple resumes against a Job Description and ranks candidates using a weighted ATS scoring engine.
- Implemented PDF text extraction, skill matching, experience and education analysis, and semantic similarity using Sentence Transformers (all-MiniLM-L6-v2) and cosine similarity.
- Designed a deterministic ATS scoring pipeline combining critical skills, supporting skills, semantic similarity, experience, and education.
- Integrated an LLM-based analysis layer to generate candidate summaries, strengths, skill gaps, and hiring recommendations.
- Developed a Streamlit dashboard supporting 15+ resumes per screening session, with candidate ranking and CSV/JSON reporting.

Full-Stack Task Management System|*Python,* React, JavaScript/JSX, PostgreSQL, SQLAlchemy, JWT— Live Demo | GitHub

- Built a full-stack task management application using **FastAPI, React, and PostgreSQL** for projects, tasks, users, and team collaboration.
- Implemented **JWT authentication, refresh tokens, password hashing, and membership-based authorization**.
- Developed a responsive **frontend using React, JavaScript, HTML, and CSS**, including dashboards, task boards, forms, comments, and project management interfaces.
- Developed task features including **assignment, priorities, status updates, comments, activity tracking, and dashboard functionality**.
- Developed **REST APIs and WebSocket-based real-time updates**, with SQLAlchemy and Alembic for database management.

EDUCATION

Brainware University — B.Tech in Computer Science & Engineering (Data Science) Kolkata, India | 2022 – 2026
CGPA: 8.57/10 | Percentage: 82%

CERTIFICATIONS

- Software Engineer Intern Certificate — HackerRank (View)
- SQL (Basic & Intermediate) Certifications — HackerRank (View)
- Artificial Intelligence — IBM SkillsBuild (View Badge)
- Python Certificate — Udemy (View)

ACHIEVEMENTS

- **HackerRank Orchestrate Hackathon 2026 participant —** Built an AI-powered financial agent for expense affordability assessment.
- Solved 200+ coding problems across LeetCode and HackerRank, covering Data Structures, Algorithms, and SQL.
- Solved 50+ SQL problems on LeetCode and earned SQL (Basic & Intermediate) badges.
- School Topper — 1st Rank in Class X (2019) and Class XII (2021).